

Hypothesis Testing with E-Values

Chapter 4: Post-hoc Testing and Decision Making

Based on Ramdas & Wang

Outline

- 1 Testing at Data-Dependent Error Levels
- 2 Post-hoc Decisions Using P-values and E-values
- 3 Optional Continuation of Experiments
- 4 Incentive-Compatible Principal-Agent Hypothesis Testing

Notation you'll need I

This chapter is about what you are allowed to do *after* you have already looked at your data.

Four ideas recur throughout; here they are in plain English before we see a single formula.

Data-dependent error level $\hat{\alpha}$

Normally you fix a significance level α (e.g. 0.05) *before* collecting data. A *data-dependent* level $\hat{\alpha}$ is a random variable (taking values in $(0,1)$) that is allowed to depend on the data in an arbitrary, even adversarial, way — for instance, you could look at how convincing your result is and only then decide how strict a bar to hold yourself to.

Post-hoc validity

A testing procedure is *post-hoc valid* if its error-control guarantee survives no matter how $\hat{\alpha}$ was chosen, even if it was chosen by a mischievous statistician trying to make the result look as significant as possible. Analogy: a post-hoc valid guarantee is like a receipt that is still honoured by the shop no matter which register you take it to, versus an ordinary p-value which is only honoured at the one specific register (the one specific α) you announced before walking in.

Notation you'll need II

Optional continuation

You ran an experiment; the evidence was suggestive but not conclusive. *Optional continuation* is the (data-dependent!) decision to collect *more* data and combine it with what you already have. The key question: can you still control your type-I error if the decision to keep going — and how much more to collect — was itself based on peeking at the first batch of data? Analogy: a detective who, after an inconclusive round of evidence-gathering, decides (based on what they've found so far) whether to keep investigating, and how.

Principal-agent setup

Two parties with different information and different incentives: a *principal* (e.g. a regulator granting licenses) who sets the rules, and an *agent* (e.g. a pharmaceutical company) who reacts to them, possibly by lying about what they believe or by running trials they know are futile. *Incentive compatibility* means the rules are designed so that agents who know their claim is false have no reason to even try.

Notation you'll need III

Recall from earlier chapters: an *e-variable* E for a null class $\mathcal{P}$ satisfies $\mathbb{E}^{\mathbb{P}}[E] \leq 1$ for every $\mathbb{P} \in \mathcal{P}$; a *p-variable* P satisfies $\mathbb{P}(P \leq \alpha) \leq \alpha$ for every $\alpha \in (0, 1)$ and every $\mathbb{P} \in \mathcal{P}$ (equivalently $\mathbb{E}^{\mathbb{P}}[\mathbb{1}\{P \leq \alpha\}] \leq \alpha$).

Why does this chapter matter?

- Standard hypothesis testing textbook theory assumes: fix α , fix your test, fix your sample size, collect data *once*, decide.
- Real science never works this way: people peek at results, change thresholds, run follow-up experiments only when the first one looks promising, and make more than one decision from a single dataset.
- **Main message of Chapter 4:** e-values are exactly the right currency for all of this — they remain valid under data-dependent significance levels, non-binary decisions, data-dependent decision problems, optional continuation, and even when the incentives of the analyst are adversarial. P-values, in general, are *not*.
- We will make every one of these claims precise, and back each one with a worked numerical example on a single running scenario: testing whether a coin is fair, $H_0 : \theta = 0.5$.

Section 4.1

- 1 Testing at Data-Dependent Error Levels
- 2 Post-hoc Decisions Using P-values and E-values
- 3 Optional Continuation of Experiments
- 4 Incentive-Compatible Principal-Agent Hypothesis Testing

Intuition: what if α is chosen after seeing the data?

- Recall: a level- α test for $\mathcal{P}$ is a function ϕ (taking values in $\{0, 1\}$, or more generally $[0, 1]$, the probability of rejecting) such that $\mathbb{E}^{\mathbb{P}}[\phi] \leq \alpha$ for every $\mathbb{P} \in \mathcal{P}$.
- As presented in earlier chapters, α was always a *predefined constant*, agreed upon before the experiment.
- **Question:** can we build a family of tests $(\phi_\alpha)_{\alpha \in (0,1)}$, one for every possible level, such that we're allowed to pick *which* α to report — using a rule $\hat{\alpha}$ that can depend on the data in *any* way — and still have $\mathbb{E}^{\mathbb{P}}[\phi_{\hat{\alpha}}] \leq \hat{\alpha}$ hold?
- This is a genuinely stronger requirement than ordinary level- α validity, because $\hat{\alpha}$ is allowed to depend on ϕ_α itself for every α simultaneously (i.e. on the whole family, not just one member).

Definition 4.1: post-hoc valid tests and p-variables I

Post-hoc valid family of tests

A collection $(\phi_\alpha)_{\alpha \in (0,1)}$ is a *post-hoc valid family of tests* for $\mathcal{P}$ if for every $\mathbb{P} \in \mathcal{P}$ and every random variable $\hat{\alpha}$ taking values in $(0, 1)$,

$$\mathbb{E}^{\mathbb{P}} \left[\frac{\phi_{\hat{\alpha}}}{\hat{\alpha}} \right] \leq 1.$$

Equivalently,

$$\mathbb{E}^{\mathbb{P}} \left[\sup_{\alpha \in (0,1)} \frac{\phi_\alpha}{\alpha} \right] \leq 1 \quad \text{for all } \mathbb{P} \in \mathcal{P}.$$

Symbols: ϕ_α is the test at nominal level α (rejection probability); $\hat{\alpha}$ is any — possibly adversarial — data-dependent level; the sup is over all possible levels at once.

Definition 4.1: post-hoc valid tests and p-variables II

Post-hoc p-variable

A nonnegative random variable P is a *post-hoc p-variable* for $\mathcal{P}$ if for every $\mathbb{P} \in \mathcal{P}$ and every random variable $\hat{\alpha} \in (0, 1)$,

$$\mathbb{E}^{\mathbb{P}} \left[\frac{\mathbb{I}\{P \leq \hat{\alpha}\}}{\hat{\alpha}} \right] \leq 1, \quad \text{equivalently} \quad \mathbb{E}^{\mathbb{P}} \left[\sup_{\alpha \in (0,1)} \frac{\mathbb{I}\{P \leq \alpha\}}{\alpha} \right] \leq 1. \quad (4.1)$$

Definition 4.1: post-hoc valid tests and p-variables III

Contrast with an ordinary p-variable. For a p-variable P , only a *weaker* condition holds:

$$\sup_{\alpha \in (0,1)} \mathbb{E}^{\mathbb{P}} \left[\frac{\mathbb{I}\{P \leq \alpha\}}{\alpha} \right] \leq 1 \quad \text{for all } \mathbb{P} \in \mathcal{P}. \quad (4.2)$$

This is precisely the usual definition of a p-variable. The crucial difference between (4.1) and (4.2) is whether the $\sup_{\alpha}$ sits *inside* or *outside* the expectation:

- (4.2): for *each fixed* α chosen in advance, the average rejection rate is $\leq \alpha$. Fine if you commit to α up front.
- (4.1): even the *worst-case, data-dependent* choice of α cannot inflate the average past 1. This is what post-hoc validity requires, and it is a fundamentally stronger property.

Definition 4.1: post-hoc valid tests and p-variables IV

Remark 4.2 (a useful generalization)

For a p-variable P , the condition $\mathbb{E}^{\mathbb{P}}[\mathbb{1}\{P \leq T\}/T] \leq 1$ (with $0/0 := 0$) also holds for some nonnegative random variables T (not necessarily in $(0, 1)$), provided

$$p \mapsto \mathbb{P}(T \leq t \mid P \leq p) \text{ is increasing for all } t \in \mathbb{R}. \quad (4.4)$$

This is a form of *negative dependence* between P and T . Examples satisfying (4.4): (a) T is a constant; (b) T is independent of P ; (c) T is a decreasing function of P .

Proposition 4.3: e-variables generate post-hoc valid families I

Proposition 4.3

Let E be an e-variable for $\mathcal{P}$. Then $((\alpha E) \wedge 1)_{\alpha \in (0,1)}$ is a post-hoc valid family of tests, and $(\mathbb{I}\{E \geq 1/\alpha\})_{\alpha \in (0,1)}$ is a post-hoc valid family of binary tests.

Conversely, every post-hoc valid family of tests is dominated by $((\alpha E) \wedge 1)_{\alpha \in (0,1)}$ for some e-variable E , and every post-hoc valid family of binary tests is dominated by $(\mathbb{I}\{E \geq 1/\alpha\})_{\alpha \in (0,1)}$ for some e-variable E .

Symbols: $x \wedge 1 = \min(x, 1)$; “dominated” means member-wise $\leq$ (a dominated family rejects no more often, hence is weakly less useful). **Punchline:** e-variables are (essentially) the *only* way to build post-hoc valid families.

Proposition 4.3: e-variables generate post-hoc valid families II

Why is this true? (proof idea) The key elementary fact used throughout this chapter is

$$\mathbb{I}\{x \geq 1\} \leq x \wedge 1 \leq x \quad \text{for } x \geq 0, \quad (4.5)$$

which is exactly the inequality underlying Markov's inequality. Applying (4.5) with $x = E/\hat{\alpha}$, for any random $\hat{\alpha} \in (0, 1)$:

$$\mathbb{E} \mathbb{P} \left[\frac{\mathbb{I}\{E \geq 1/\hat{\alpha}\}}{\hat{\alpha}} \right] \leq \mathbb{E} \mathbb{P} \left[\frac{(\hat{\alpha}E) \wedge 1}{\hat{\alpha}} \right] \leq \mathbb{E} \mathbb{P} \left[\frac{\hat{\alpha}E}{\hat{\alpha}} \right] = \mathbb{E} \mathbb{P}[E] \leq 1.$$

This one chain of inequalities is the engine behind almost every result in this section: it shows both families are post-hoc valid *for any* $\hat{\alpha}$ whatsoever, adversarial or not, because $\hat{\alpha}$ simply cancels out.

Converse (sketch): given a post-hoc valid family (ϕ_α) , set $E = \sup_\alpha \phi_\alpha/\alpha$; post-hoc validity says exactly that E is an e-variable, and by construction $\phi_\alpha \leq (\alpha E) \wedge 1$.

Proposition 4.4: post-hoc p-variables are (essentially) $1/E$

Proposition 4.4

For a random variable P , the following are equivalent:

- 1 P is a post-hoc p-variable for $\mathcal{P}$;
- 2 $P = 1/E$ on the event $\{P < 1\}$, for some e-variable E for $\mathcal{P}$;
- 3 $P \geq (1/E) \wedge 1$ for some e-variable E for $\mathcal{P}$.

If instead we strengthen (4.1) by letting α range over $(0, \infty)$,

$$\sup_{\alpha \in (0, \infty)} \mathbb{E}^{\mathbb{P}} \left[\frac{\mathbb{1}_{\{P \leq \alpha\}}}{\alpha} \right] \leq 1 \quad \text{for all } \mathbb{P} \in \mathcal{P}, \quad (4.6)$$

then a post-hoc p-variable is *precisely* $P = 1/E$ for some e-variable E (which may exceed 1, since we allow p-variables larger than 1). **Takeaway:** post-hoc p-variables and e-variables are really the same object, viewed from two sides of the same coin: $P = 1/E$.

Interpreting the stronger guarantee: $E = 41$ vs. $P = 0.024$

Suppose on the same hypothesis we observe an e-value $E \approx 41.05$ or a p-value $P \approx 0.0244$ on the same data (as we compute concretely below). These look “comparable” ($1/E \approx P$), but they are *not* interchangeable:

- Because of Proposition 4.3–4.4, we can *safely* say: “we reject H_0 at level 0.0244 (chosen after seeing E !), and even at level 0.05.” This claim is protected by post-hoc validity.
- We *cannot* safely say: “we reject H_0 at level 0.05” based on $P \approx 0.0244$, *unless the level 0.05 was fixed before collecting and inspecting the data*.
- If the level was chosen only *after* peeking (e.g. “the p-value looked good, so let’s call 0.05 our threshold”), the ordinary p-value’s guarantee (4.2) says nothing, because (4.2) never promised anything about a $\hat{\alpha}$ that depends on the data.

Rule of thumb: an e-value of $1/a$ carries *strictly more* information than a p-value of a , precisely because of this extra post-hoc guarantee.

Running example, Part 1: peeking is safe with e-values

Setup. $H_0 : \theta = 0.5$ (fair coin). A scientist tosses the coin $n = 20$ times and forms the likelihood-ratio e-value against the simple alternative $\theta = 0.75$:

$$E(x) = \left(\frac{0.75}{0.5}\right)^H \left(\frac{0.25}{0.5}\right)^T = 1.5^H \cdot 0.5^T,$$

where H = number of heads, $T = 20 - H$ = number of tails. This E is an e-variable for H_0 : $\mathbb{E}^{\theta=0.5}[E] = 1$ exactly (it's a likelihood ratio).

Data. Suppose she observes $H = 16$ heads, $T = 4$ tails. Then

$$E = 1.5^{16} \times 0.5^4 \approx 41.05.$$

Post-hoc decision. After seeing $E \approx 41.05$, she decides — on the spot, informed by how good the number looks — to report significance at the data-dependent level

$$\hat{\alpha} = 1/E \approx 0.0244$$

(the smallest level her evidence supports). By Proposition 4.3–4.4 this is completely valid: the family $(\mathbb{I}\{E \geq 1/\alpha\})_\alpha$ is post-hoc valid *for any* rule choosing $\hat{\alpha}$, including this self-serving one. She may honestly report “significant at level 0.0244, chosen after seeing the data.”

Running example, Part 2: peeking breaks p-value validity

Same idea, now with an ordinary p-value. Under H_0 , suppose P is (as usual) continuous with $P \sim \text{Uniform}(0, 1)$.

The adversarial choice. Consider a scientist who, having computed P , sets her “after-the-fact” threshold to be exactly her own p-value: $\hat{\alpha} := P$. Then trivially

$$\mathbb{P}(P \leq \hat{\alpha}) = \mathbb{P}(P \leq P) = 1,$$

i.e. she can *always* claim “significant at level $\hat{\alpha}$ ” — a 100% false-rejection rate under the null, however small α “looks”! Compare to a pre-registered $\alpha = 0.05$: type-I error is exactly 0.05, as it should be.

Same fact, via Definition 4.1. The quantity that post-hoc validity requires to be ≤ 1 blows up for p-values:

$$\mathbb{E} \mathbb{P} \left[\sup_{\alpha \in (0,1)} \frac{\mathbb{1}\{P \leq \alpha\}}{\alpha} \right] = \mathbb{E} \mathbb{P} \left[\frac{1}{P} \right] = \int_0^1 \frac{1}{p} dp = \infty.$$

Contrast: for the e-value family, the analogous quantity is $\mathbb{E} \mathbb{P}[E \cdot \mathbb{1}\{E \geq 1\}] \leq \mathbb{E} \mathbb{P}[E] \leq 1$ — finite, by Markov’s inequality, no matter what. This is the concrete, numerical face of “post-hoc validity for e-values, but not for p-values.”

Python: post-hoc risk – e-values vs. p-values (§4.1)

```
import numpy as np
from scipy.special import erfc

rng = np.random.default_rng(0)
theta0, theta1, n, M = 0.5, 0.6, 30, 200_000

# simulate under H0: theta = 0.5
heads = rng.binomial(n, theta0, size=M)
tails = n - heads

# likelihood-ratio e-value for theta1 vs theta0
E = (theta1/theta0)**heads * ((1-theta1)/(1-theta0))**tails

# continuous one-sided p-value (normal approximation)
z = (heads - n*theta0) / np.sqrt(n*theta0*(1-theta0))
P = np.clip(0.5*erfc(z/np.sqrt(2)), 1e-300, 1.0)

R_P = 1.0 / P                                # sup_alpha 1{P<=alpha}/alpha
R_E = E * (E >= 1.0)                         # sup_alpha 1{E>=1/alpha}/alpha

print("running avg R_P (p-values):", np.mean(R_P))    # unstable, >> 1
print("running avg R_E (e-values):", np.mean(R_E))    # stays <= 1
```

Output (M=200,000 replicates): running average of $R_P \approx 15$ (unstable, keeps growing with M since $\mathbb{E}[1/P] = \infty$), while running average of $R_E \approx 0.71 \leq 1$, exactly as guaranteed by Markov's inequality applied to the e-variable E . See figures/fig_posthoc_validity.pdf for the full trajectory.

Section 4.1 take-aways

- Ordinary level- α tests only control error *for a fixed, pre-registered* α .
- Post-hoc validity (Def. 4.1) additionally controls error for *any*, even adversarially data-dependent, choice of level $\hat{\alpha}$.
- e-variables generate post-hoc valid families essentially uniquely (Prop. 4.3), and post-hoc p-variables are just $1/E$ (Prop. 4.4).
- Concretely: it is fine to look at your e-value and then decide how impressive a claim to make; doing the analogous thing with an ordinary p-value can inflate your true error rate all the way to 100%.

Section 4.2

- 1 Testing at Data-Dependent Error Levels
- 2 Post-hoc Decisions Using P-values and E-values
- 3 Optional Continuation of Experiments
- 4 Incentive-Compatible Principal-Agent Hypothesis Testing

Intuition: beyond accept/reject

- So far, “decisions” meant a binary choice: reject H_0 or not.
- Real decision-making is richer: a doctor may choose among *several* treatment intensities; a regulator may choose among several licensing tiers; and *which* decisions are even on the table may itself depend on what the data show (a data-dependent “scenario”).
- Section 4.2 asks: if we build our decision rule from a p-value vs. from an e-value, which one keeps its risk-control promise once decisions become non-binary, or once the decision problem itself depends on data?
- **Spoiler:** for the simplest case (binary decisions, fixed scenario), both work. As soon as either non-binary decisions or data-dependent scenarios enter, only the e-value-based rule survives.

Setup: loss functions and the risk-constrained problem I

Consider loss functions $L_b : \{0, 1\} \times D_b \rightarrow \mathbb{R}$ indexed by a *scenario* $b \in \mathcal{B} \subseteq \mathbb{R}$, where $D_b \subseteq \mathbb{N}_0$ is the set of possible decisions in scenario b .

- $L_b(0, d)$: loss to the decision maker from decision d *when H_0 is true*;
- $L_b(1, d)$: loss from decision d *when the alternative is true*;
- WLOG, $d \mapsto L_b(0, d)$ is strictly increasing and $d \mapsto L_b(1, d)$ is strictly decreasing — larger d = a “more aggressive” decision: costlier if H_0 true, more beneficial if the alternative is true.

Data $X \in \mathcal{Z}$; decision rule $\delta(X) \in D_b$.

Risk-constrained decision problem (4.7)

$$\text{minimize}_{\delta: \mathcal{Z} \rightarrow D_b} \mathbb{E}^{\mathbb{Q}}[L_b(1, \delta(X))] \quad \text{s.t.} \quad \sup_{\mathbb{P} \in \mathcal{P}} \mathbb{E}^{\mathbb{P}}[L_b(0, \delta(X))] \leq \gamma,$$

where $\gamma > 0$ is a prespecified *risk budget*, and the constraint is the *risk constraint*. (We fix a simple alternative $\mathcal{Q} = \{\mathbb{Q}\}$ for simplicity; the composite case is analogous.)

Example 4.5: ternary significance decisions

A very common real practice: label a result “non-significant (NS)”, “significant (S)”, or “very significant (VS)” after seeing the data.

$$D_b = \{0, 1, 2\}, \quad 0 = \text{NS}, \quad 1 = \text{S}, \quad 2 = \text{VS}.$$

If H_0 true, decision 0 (NS) has least loss; if the alternative is true, decision 2 (VS) has least loss. A common choice:

$$\gamma = 1, \quad L_b(0, 0) = 0, \quad L_b(0, 1) = \frac{1}{\alpha_1}, \quad L_b(0, 2) = \frac{1}{\alpha_2},$$

for some $0 < \alpha_2 < \alpha_1 < 1$ (e.g. $\alpha_1 = 0.05$, $\alpha_2 = 0.01$).

Decision rules based on p-values and e-values

Suppose a p-variable $P(X)$ and an e-variable $E(X)$ for $\mathcal{P}$ are both computed from the data.

p-value decision rule (4.8)

$$\delta_P(X) = \max\{d \in D_b : P(X) L_b(0, d) \leq \gamma\}.$$

e-value decision rule (4.9)

$$\delta_E(X) = \max\{d \in D_b : L_b(0, d) \leq \gamma E(X)\}.$$

Symbols: both rules pick the most aggressive decision d still compatible with the risk budget γ , using either $P(X)$ or $E(X)$ as the “evidence multiplier.”

Example 4.6: ternary decisions, worked out with numbers

Take $\alpha_1 = 0.05$, $\alpha_2 = 0.01$, $\gamma = 1$ (as in Example 4.5).

p-value rule (4.8)

$$\delta_P(X) = \begin{cases} 2 & \text{if } P(X) \leq 0.01, \\ 1 & \text{if } 0.01 < P(X) \leq 0.05, \\ 0 & \text{otherwise.} \end{cases}$$

This is standard scientific practice (“*” vs. “**” asterisks on p-values) – **but it will turn out not to satisfy the risk constraint (4.7) in general (Example 4.10)!**

e-value rule (4.9)

$$\delta_E(X) = \begin{cases} 2 & \text{if } E(X) \geq 100, \\ 1 & \text{if } 20 \leq E(X) < 100, \\ 0 & \text{otherwise.} \end{cases}$$

This rule *always* satisfies the risk constraint (Theorem 4.9).

Binary decisions in a fixed scenario: Proposition 4.7 I

Simplest case: $D_b = \{0, 1\}$ (0=accept, 1=reject), and WLOG $L_b(0, 0) = L_b(1, 1) = 0$, $L_b(0, 1) = L_b(1, 0) = 1$. Then (4.7) becomes the familiar

$$\text{maximize } \mathbb{Q}(\delta(X) = 1) \quad \text{s.t.} \quad \sup_{\mathbb{P} \in \mathcal{P}} \mathbb{P}(\delta(X) = 1) \leq \gamma, \quad (4.10)$$

i.e. maximize power subject to type-I error control γ .

- p-value rule: $\delta_P(X) = 1$ iff $P(X) \leq \gamma$.
- e-value rule: $\delta_E(X) = 1$ iff $E(X) \geq 1/\gamma$.

Proposition 4.7

Given a fixed scenario b with $D_b = \{0, 1\}$, both δ_P and δ_E satisfy the risk constraint in (4.10).

Binary decisions in a fixed scenario: Proposition 4.7 II

Proof. For any $\mathbb{P} \in \mathcal{P}$:

$$\mathbb{E}^{\mathbb{P}}[L_b(0, \delta_P(X))] = \mathbb{P}(\delta_P(X) = 1) = \mathbb{P}(P(X) \leq \gamma) \leq \gamma,$$

using the p-variable property. And

$$\mathbb{E}^{\mathbb{P}}[L_b(0, \delta_E(X))] \leq \mathbb{E}^{\mathbb{P}}[\gamma E(X)] \leq \gamma,$$

using $L_b(0, \delta_E(X)) \leq \gamma E(X)$ by construction, and $\mathbb{E}^{\mathbb{P}}[E] \leq 1$.

Conclusion for this simple case: both approaches are valid! Since $1/E(X)$ is itself a p-variable (Section 2.1), the e-value rule is typically *more conservative* than the p-value rule here. The real difference emerges only once we leave this simplest binary/fixed-scenario world.

Non-binary or data-dependent scenarios: Example 4.8

Vaccine story. A clinical trial's data can put the scientist in one of two *scenarios* $B \in \{1, 2\}$:

- $B = 1$ (“promising”): decisions $D_1 = \{0, 1\}$, where $\delta(X, 1) = 0$ means *don't vaccinate anyone* and $\delta(X, 1) = 1$ means *vaccinate the highest-risk population*.
- $B = 2$ (“extremely effective”, unexpected): now $\delta(X, 2) = 1$ means something more aggressive: *vaccinate the full population*.

Toy numeric loss table

Scenario b	$L_b(0, 0)$	$L_b(0, 1)$	meaning of $d = 1$
$B = 1$ (promising)	0	1	vaccinate high-risk only
$B = 2$ (extremely effective)	0	10	vaccinate everyone

Note D_1 and D_2 carry *different meanings* for the same label $d = 1$ – and which scenario B you're even in is itself random, determined by the data!

The formal data-dependent problems (4.11)-(4.12) I

Write B (a function of X) for the observable, data-dependent scenario; the decision rule now takes both X and B : $\delta(X, B) \in D_B$.

Problem (4.11): scenario B known to the decision maker

$$\text{minimize } \mathbb{E}^{\mathbb{Q}}[L_B(1, \delta(X, B))] \quad \text{s.t.} \quad \sup_{\mathbb{P} \in \mathcal{P}} \mathbb{E}^{\mathbb{P}}[L_B(0, \delta(X, B))] \leq \gamma.$$

Problem (4.12): worst-case objective/constraint over all $b \in \mathcal{B}$

$$\text{minimize } \mathbb{E}^{\mathbb{Q}}\left[\sup_{b \in \mathcal{B}} L_b(1, \delta(X, b))\right] \quad \text{s.t.} \quad \sup_{\mathbb{P} \in \mathcal{P}} \mathbb{E}^{\mathbb{P}}\left[\sup_{b \in \mathcal{B}} L_b(0, \delta(X, b))\right] \leq \gamma.$$

The e-value rule extends naturally to this setting and by construction satisfies, for every b ,

$$L_b(0, \delta_E(X, b)) \leq \gamma E(X). \tag{4.13}$$

The formal data-dependent problems (4.11)-(4.12) II

Theorem 4.9

- ① δ_E (for any e-variable $E(X)$) satisfies the risk constraints in both (4.11) and (4.12).
- ② Any δ^* satisfying the constraint in (4.11) can be written as $\delta_E(X, B)$ for some e-variable $E(X)$.
- ③ Any δ^* satisfying the constraint in (4.12) satisfies $\delta^*(X, b) \leq \delta_E(X, b)$ for some e-variable $E(X)$ and all $b \in \mathcal{B}$ (i.e. δ^* is weakly improved by using e-variables).
- ④ The p-value rule δ_P does *not* satisfy the risk constraint in (4.11) or (4.12) in general, once D_b has at least 3 elements or $\mathcal{B}$ has at least 2 elements.

Proof of (i): directly from (4.13), $\mathbb{E}^{\mathbb{P}}[\sup_b L_b(0, \delta_E(X, b))] \leq \mathbb{E}^{\mathbb{P}}[\gamma E(X)] \leq \gamma$.

Example 4.10: a concrete counterexample for δ_P

Let $P(X) \sim \text{Uniform}[0, 1]$ under $\mathbb{P} \in \mathcal{P}$. Take $\mathcal{B} = \{1, 2\}$, $D_b = \{0, 1\}$, $L_1(0, 1) = \ell_1$, $L_2(0, 1) = \ell_2$ with $\ell_1 < \ell_2$, and let the scenario be $B = 2$ exactly when $P(X) \leq \gamma/\ell_2$ (otherwise $B = 1$). A direct computation gives

$$\mathbb{E}^{\mathbb{P}}[L_B(0, \delta_P(X))] = \ell_1 \mathbb{P}\left(\frac{\gamma}{\ell_2} \leq P(X) \leq \frac{\gamma}{\ell_1}\right) + \ell_2 \mathbb{P}\left(P(X) \leq \frac{\gamma}{\ell_2}\right) = \gamma\left(1 - \frac{\ell_1}{\ell_2}\right) + \gamma > \gamma. \quad (4.14)$$

Concrete numbers: take $\gamma = 0.05$, $\ell_1 = 1$, $\ell_2 = 10$. Then the achieved risk is

$$0.05\left(1 - \frac{1}{10}\right) + 0.05 = 0.045 + 0.05 = 0.095,$$

nearly double the allowed budget $\gamma = 0.05$! The p-value rule looks fine *within* each fixed scenario, but silently breaks the overall risk guarantee once the scenario itself is chosen using the data. By contrast (Theorem 4.9(i)), δ_E never does this.

Python: verifying Example 4.10 by simulation (§4.2)

```
import numpy as np
rng = np.random.default_rng(1)

gamma, ell1, ell2 = 0.05, 1.0, 10.0
M = 2_000_000
P = rng.uniform(0, 1, size=M)          # P-variable under H0

# scenario B depends on the data:
B = np.where(P <= gamma/ell2, 2, 1)

# delta_P(X) = 1 iff P(X) * L_B(0,1) <= gamma
L_B_of_1 = np.where(B == 2, ell2, ell1)
delta_P = (P * L_B_of_1 <= gamma).astype(int)

loss = np.where(delta_P == 1, L_B_of_1, 0.0)
print("empirical risk of delta_P:", loss.mean(), " (budget =", gamma, ")")
# --> approx 0.095, almost double the risk budget gamma = 0.05
```

The simulated risk matches the closed-form value 0.095 from (4.14), confirming that a p-value-based ternary/scenario-dependent rule can silently blow through its risk budget.

Section 4.2 take-aways

- With *binary* decisions and a *fixed* scenario, both p-value- and e-value-based decision rules are valid (Prop. 4.7); e-value rules tend to be more conservative here.
- As soon as decisions are non-binary (e.g. NS/S/VS) *or* the decision problem itself is data-dependent (the scenario B depends on X), only e-value-based rules keep the risk-control promise (Theorem 4.9); p-value-based rules can silently violate it (Example 4.10).
- Practical warning: this failure mode is common and often *unconscious* – e.g. marking p-values with significance asterisks after the fact is exactly this ternary-decision setting.

Section 4.3

- 1 Testing at Data-Dependent Error Levels
- 2 Post-hoc Decisions Using P-values and E-values
- 3 Optional Continuation of Experiments**
- 4 Incentive-Compatible Principal-Agent Hypothesis Testing

Intuition: “I should have collected more data”

- A scientist runs a carefully pre-planned experiment. The result is close, but not quite significant. In hindsight, they realize they underestimated the effect size and should have collected more data.
- With a p-value, this realization is *useless*: the type-I error budget for that test was already spent the moment they committed to that sample size and threshold. Tacking on more data and re-testing (even honestly) inflates the false-positive rate.
- **The question of Section 4.3:** is there a principled way to let a scientist decide — *after seeing the first batch of data* — whether to run more, and if so, keep a valid guarantee against H_0 ? We call this *optional continuation*.
- The answer, again, is yes with e-values.

Coin-fairness example: the Neyman–Pearson baseline

Setup. Test $H_0 : \theta = 0.5$ (fair coin) vs. an anticipated bias $\theta = 0.6$. To guarantee both type-I and type-II error at 0.05, the optimal (Neyman–Pearson) likelihood-ratio test:

- tosses the coin $n = 280$ times;
- rejects H_0 iff at least 154 heads appear.

Suppose, on performing those 280 tosses, the scientist does *not* reject. Maybe it's a type-II error; maybe they should have planned for a smaller bias, say $\theta = 0.55$. **But it's too late:** the type-I error budget of this test has already been fully spent, and re-examining the same data with a new threshold is not valid. With p-values, such regrets are simply futile.

E-values to the rescue: combining fresh batches I

Rewrite the original test as: reject when the LR-test p-value drops below 0.05. Now instead summarize the very same 280-toss experiment as an e-value E_1 from the start (the likelihood ratio is a natural choice).

Key martingale property

Suppose the scientist (or someone else) is permitted to toss the coin more times – say 50 more – and computes a fresh e-value E_2 based *only* on this new batch. Under the null,

$$\mathbb{E}^{\mathbb{P}}[E_2 \mid E_1] = 1.$$

Then $E^{(2)} := E_2 E_1$ is *also* a valid e-value for the same null (by the law of iterated expectation: $\mathbb{E}^{\mathbb{P}}[E^{(2)}] = \mathbb{E}^{\mathbb{P}}[\mathbb{E}^{\mathbb{P}}[E_2 E_1 \mid E_1]] = \mathbb{E}^{\mathbb{P}}[E_1 \cdot 1] = 1$). Crucially: *both* the decision to run this second experiment *and* its design (e.g. how many more tosses) are allowed to depend on the first batch of data – this is a post-hoc decision, and it remains valid!

E-values to the rescue: combining fresh batches II

The scientist may still be unsatisfied. They can collect a third batch, summarize it as E_3 , and form $E^{(3)} := E_3 E^{(2)}$, itself a valid e-value. **This can be repeated indefinitely**: the evidence process $E^{(1)}, E^{(2)}, \dots$ can be monitored continually, and H_0 rejected as soon as the current evidence exceeds $1/\alpha$ – this is controlled by *Ville's inequality* (Chapter 7). Alternatively, the scientist may stop at any data-dependent stopping time τ and report $E^{(\tau)}$, itself an e-value, by the *optional stopping theorem* (Chapter 7). When each batch shrinks to a single toss, this becomes a fully sequential test – Wald's SPRT (Section 7.2).

Worked numbers: from “promising” to a valid rejection

Continuing our coin example. Suppose the scientist’s original $n = 280$ -toss batch yields e-value

$$E_1 = 12.$$

Since we need $E \geq 1/\alpha = 20$ to reject at $\alpha = 0.05$, they cannot reject yet – but the post-hoc p-value $1/E_1 \approx 0.083$ is “promising.” *Because* it looked promising, they decide (post-hoc!) to run 50 more tosses, obtaining a fresh e-value on the new batch alone:

$$E_2 = 2.5.$$

The combined e-value is

$$E^{(2)} = E_1 \times E_2 = 12 \times 2.5 = 30 \geq 20,$$

so **they may now validly reject H_0 at level $\alpha = 0.05$** – even though the second experiment was only run *because* the first one was almost significant, and even though this is exactly the kind of peeking that would be illegitimate with p-values.

Optional continuation vs. optional stopping: who's continuing?

- This setting is intimately related to sequential testing / optional stopping (Chapter 7), but is *motivated differently*.
- Here, the scientist who continues the study may be a *different* person from the one who collected the first batch. Data collection may be expensive – the first scientist may simply have run out of money, or given up because the evidence looked unpromising.
- We call this setting *optional continuation*, in contrast to *optional stopping* (Chapter 7), where a *single* experimenter controls if/when to stop a single ongoing experiment.
- Mathematically the multiplicative combination rule is identical; the distinction is about *who* is making the post-hoc decision, and why.

The “living meta-analysis” story

Scenario. A drug (e.g. a pandemic vaccine) is expensive to test.

- Country A runs an expensive trial; the findings are impressive but Country B still deems it too risky to launch without its own evidence.
- Country B, piggy-backing on Country A's success, runs its own trial and *appends* its e-value evidence to Country A's.
- Country C, who wasn't even considering the drug, may now be convinced to run its own trial too.

As long as every country's evidence is summarized as an e-value, the overall product $E^{(1)}E^{(2)}E^{(3)}\dots$ is easy to update and monitor sequentially across this unplanned, ad hoc sequence of trials. **Toy numbers:** wealthier countries might launch a vaccine program once the combined e-value exceeds 20 (i.e. post-hoc level ≤ 0.05), while poorer, more risk-averse countries wait until it exceeds 100 (post-hoc level ≤ 0.01) before acting *without even running their own trial*.

Three important caveats I

1. No hiding unfavorable findings

No participant may discard or hide their result, even if their own batch gives a small e-value (less than 1, i.e. evidence *for* H_0). Once the decision to run the experiment is made, the resulting e-value *must* be incorporated into the combined evidence.

2. The “poison pill” problem, and two fixes

A single scientist reporting an e-value of exactly 0 can permanently destroy *all* previously accumulated evidence (since the running product becomes 0 forever after). Two ways to guard against this:

- Agree in advance to use e-values that are never exactly 0 by design (e.g. log-optimal e-values);
- Or agree to combine via $\prod_{i=1}^n \left(\frac{1+E_i}{2}\right)$, so no single batch can destroy more than half of the existing evidence – generalizable to $\prod_{i=1}^n (\lambda_i + (1 - \lambda_i)E_i)$ with $\lambda_i \in [0, 1]$ *fixed before seeing* E_i .

Either way: the i -th e-value's construction, and how it is combined with the previous ones, must be specified *before* seeing the i -th batch of data.

Three important caveats II

3. The decision to continue cannot be based on the p-value “not being small enough”

The decision to continue collecting data at a second stage *cannot* be triggered by “the first p-value wasn’t quite below 0.05” – testing at the first stage with that p-value already used up the entire type-I error budget. One must commit either to using a p-value with a single, fixed batch, *or* to using an e-value with the possibility of extending the analysis – not switch between the two after the fact.

Why can't we just combine p-values the same way?

It is tempting to think the second batch (say Country B's trial data) is “independent” of the first, and to merge the two p-values with a standard rule for independent p-values (e.g. Fisher's method). **This would be a mistake.**

- The second p-value is *not* really independent of the first, despite being computed on “separate” data: it would not even exist had the first p-value not looked promising enough to justify continuing!
- So the *number* of p-values collected is itself random, and the p-values are *sequentially dependent* in a complicated way that standard combination rules do not account for.
- Such post-hoc decisions to run further trials, or to fold in accumulated data from separate trials, are a defining feature of e-value-based evidence collection – and a fundamental obstacle for p-value-based analysis.

Python: e-process vs. naive p-value peeking (§4.3)

```
import numpy as np
from scipy.special import erfc
rng = np.random.default_rng(2)

theta0, theta1 = 0.5, 0.6
n_batches, batch_size, n_runs, alpha = 20, 10, 20_000, 0.05
threshold = 1/alpha # = 20

heads = rng.binomial(batch_size, theta0, size=(n_runs, n_batches))
tails = batch_size - heads

e_batch = (theta1/theta0)**heads * ((1-theta1)/(1-theta0))**tails
z = (heads - batch_size*theta0)/np.sqrt(batch_size*theta0*(1-theta0))
p_batch = np.clip(0.5*erfc(z/np.sqrt(2)), 1e-300, 1.0)

e_process = np.cumprod(e_batch, axis=1) # the e-process
ever_reject_e = (np.cumsum(e_process >= threshold, axis=1) > 0)
ever_reject_p = (np.cumsum(p_batch < alpha, axis=1) > 0) # naive peeking

print("false-rejection rate, e-process (Ville):", ever_reject_e[:, -1].mean())
print("false-rejection rate, naive p peeking :", ever_reject_p[:, -1].mean())
```

Output (20 looks, $\alpha = 0.05$): e-process monitoring stays at $\approx 0.028 \leq 0.05$ (Ville's inequality holds), while naive repeated p-value peeking inflates to ≈ 0.675 – more than *thirteen times* the nominal level. See figures/fig_optional_continuation.pdf.

Section 4.3 take-aways

- E-values multiply: fresh e-values on fresh (even post-hoc decided) batches combine into a valid e-value for the whole history, by the tower property $\mathbb{E}^{\mathbb{P}}[E_2 \mid E_1] = 1$.
- This licenses *optional continuation*: deciding whether, and how, to gather more data *after* seeing preliminary results — something ordinary p-values cannot support once the first test's error budget is spent.
- Three caveats keep the guarantee honest: no hiding results, guard against a poison-pill e-value of 0, and never let the p-value's own "closeness to significance" trigger continuation.
- Sequential monitoring of the e-process is controlled by Ville's inequality (previewed here, developed fully in Chapter 7).

Section 4.4

- 1 Testing at Data-Dependent Error Levels
- 2 Post-hoc Decisions Using P-values and E-values
- 3 Optional Continuation of Experiments
- 4 Incentive-Compatible Principal-Agent Hypothesis Testing

Intuition: the regulator and the pharma company

- Two parties: a *principal* (e.g. a regulator granting drug licenses) and an *agent* (e.g. a pharmaceutical company running a trial).
- Two features make this different from ordinary testing:
 - ① **Information asymmetry**: the agent may know far more about whether their drug actually works than the regulator does.
 - ② **Conflicting incentives**: the regulator wants to keep ineffective drugs off the market; the agent wants a license to sell and profit, whether or not the drug works.
- Standard practice: if $P < 0.05$, grant the license. We will see *numerically* why this is not *incentive compatible*, and how e-values fix it.

The incentive-compatibility problem: a numeric payoff table

A testing paradigm is *incentive compatible* if it discourages *malicious* agents (who know H_0 is true, e.g. they have a placebo) from even attempting a trial. Suppose a trial costs \$10 million, and the standard rule is: license granted iff $P < 0.05$.

Toy payoff table for a malicious agent (drug is a placebo)

Anticipated profit if approved	Trial cost	Pr(approval H_0)	Expected net profit
\$100 million	\$10 million	0.05	$100(0.05) - 10 = -\$5\text{M}$
\$1,000 million (\$1B)	\$10 million	0.05	$1000(0.05) - 10 = +\$40\text{M}$

Reading the table: if the anticipated profit is modest (\$100M), the malicious agent expects to *lose* money and won't run the trial – good. But if the anticipated profit is large enough (\$1B), the malicious agent's expected profit is strongly *positive*: they will run the trial *purely to gamble on the 5% chance of approval*. The threshold rule is *not* incentive compatible once profits are large enough.

Why this matters at scale

A quick back-of-envelope calculation

Suppose there are 20 times as many ineffective drugs (placebos, in development) as truly effective ones, and 5% of ineffective drugs get approved (the false-positive rate $\alpha = 0.05$).

Drug type	Relative count	Approval rate	Expected # approved
Ineffective	20	5%	$20 \times 0.05 = 1.0$
Effective	1	$\leq 100\%$ (power)	≤ 1.0

Since the expected number of *ineffective* drugs approved (1.0) is already at least as large as the expected number of *effective* drugs approved (≤ 1.0), **at least half of all approved drugs will be ineffective** – a stark illustration of why the naive threshold-on-a-p-value protocol is dangerous at a population level, quite apart from the incentive problem above.

Designing a better protocol

Another problem with the naive protocol: a malicious agent's potential profit is limited only by *market size*, not by the strength of their statistical evidence. We would like a protocol with two properties:

- 1 **Incentive compatible:** malicious agents have no incentive to participate, regardless of market size.
- 2 **Evidence-scaled reward:** if an agent participates, their permitted sales/profit should *increase with the strength of the statistical evidence* for effectiveness.

We will see that both are easily achieved with a *statistical contract* built from e-values.

The statistical contract

Let $\theta \in \Theta$ be the drug's true quality (known to the agent), with $\Theta_0 \subsetneq \Theta$ the “placebo” parameter values the regulator wants to keep off the market.

Protocol: the agent follows these steps, in order

- 1 Choose a nonnegative *license function* f from a menu of options $\mathcal{F}$ (offered by the regulator).
- 2 Incur a fixed cost C to run a trial, collecting data $Z \stackrel{d}{\sim} \mathbb{P}_\theta$.
- 3 Receive a license to profit at most $L = f(Z)$.

The agent's expected profit is $\mathbb{E}^{\mathbb{P}_\theta}[f(Z)] - C$; they participate iff this is positive. The regulator wants this to be *negative* whenever $\theta \in \Theta_0$ (a placebo).

Proposition 4.11: characterizing incentive compatibility

Proposition 4.11

The statistical contract is incentive compatible if and only if $f(Z)/C$ is an e-variable for $\{\mathbb{P}_\theta : \theta \in \Theta_0\}$, for every $f \in \mathcal{F}$.

Proof. Incentive compatible means $\mathbb{E}^{\mathbb{P}_\theta}[f(Z)] \leq C$ for all $\theta \in \Theta_0$, i.e. $\mathbb{E}^{\mathbb{P}_\theta}[f(Z)/C] \leq 1$ for all $\theta \in \Theta_0$ – precisely the definition of $f(Z)/C$ being an e-variable for the placebo class. ■

This is exactly Section 4.2's decision-maker framework, with the regulator constructing a menu of *scaled e-values* as the allowed license functions.

A p-to-e calibrator fixes the naive protocol

Suppose evidence is summarized as a p-value P . It might be tempting to promise a license with profit margin *inversely proportional* to P , i.e. $f(Z) = C/P$.

- This is **not** incentive compatible: since $\mathbb{E}^{\mathbb{P}_\theta}[1/P] = \infty$ (as we computed in Section 4.1!), the expected profit under the null is *infinite*.
- Instead, awarding a license of

$$f(Z) = \frac{C}{2\sqrt{P}}$$

is incentive compatible: $p \mapsto 1/(2\sqrt{p})$ is a *p-to-e calibrator*, i.e. $\mathbb{E}^{\mathbb{P}_\theta}[1/(2\sqrt{P})] \leq 1$ under H_0 , so $\mathbb{E}^{\mathbb{P}_\theta}[f(Z)] \leq C$ for all $\theta \in \Theta_0$ as required.

- Intuition: the calibrator “tempers” the explosive $1/P$ tail just enough to keep its expectation finite – turning a p-value into a genuine e-value.

Python: checking incentive compatibility of license functions (§4.4)

```
import numpy as np
rng = np.random.default_rng(3)

M = 2_000_000
P = rng.uniform(0, 1, size=M)      # p-value under a placebo (H0 true)
C = 10.0                          # trial cost, in $ millions

f_naive = C / P                    # naive: profit inversely prop. to P
f_calibrated = C / (2*np.sqrt(P)) # p-to-e calibrator: 1/(2 sqrt(p))

print("E[f_naive(Z)]/C      :", np.mean(f_naive)/C, " (theory: infinite)")
print("E[f_calibrated(Z)]/C :", np.mean(f_calibrated)/C, " (theory: <= 1)")
```

Empirically, f_{naive} keeps growing without bound as M increases (occasional tiny P dominate the average), confirming it is *not* incentive compatible; $f_{\text{calibrated}}$ stabilizes at a value ≤ 1 , confirming $f(Z)/C$ is a genuine e-variable.

Connecting back to Section 4.2: two decision makers

- 1 **The regulator moves first:** they have no objective to maximize, but must meet the constraint of incentive compatibility. This is achieved by offering a menu $\mathcal{F}$ composed of (scaled) e-values; to encourage effective drugs to participate, $\mathcal{F}$ should otherwise be *unrestricted* – the set of *all* scaled e-values.
- 2 **The agent moves second:** facing a data-dependent loss (their payout depends on the data they will collect), and with incentive compatibility already guaranteed by the regulator's choice of $\mathcal{F}$, their decision is simple: participate iff some $f \in \mathcal{F}$ is profitable in expectation under their true (privately known) θ . Malicious agents (who know $\theta \in \Theta_0$) never will; agents with genuinely effective drugs may.

Summary: smart use of data-dependent loss functions, built from a menu of scaled e-values, considerably simplifies the decision problem for *both* parties.

Section 4.4 take-aways

- Fixed p-value thresholds are not incentive compatible: a malicious agent with a large enough potential market will always find it profitable to gamble on the false-positive rate.
- Incentive compatibility is exactly characterized (Prop. 4.11) by requiring the license/profit function, scaled by the trial cost, to be an e-variable for the “placebo” null class.
- A p-to-e calibrator (e.g. $1/(2\sqrt{P})$) converts a naive, incentive-*in*compatible p-value-based reward into an incentive compatible one.
- This is the same decision-theoretic machinery as Section 4.2, applied to a two-player (principal-agent) game instead of a single decision maker.

Chapter 4: the big picture

- **4.1** Post-hoc validity: e-values (uniquely) support choosing your significance level *after* seeing the data, with a guarantee that survives even adversarial choices of $\hat{\alpha}$. Ordinary p-values do not: $\mathbb{E}[1/P] = \infty$.
- **4.2** Post-hoc decisions: e-value-based decision rules satisfy risk constraints even for non-binary decisions and data-dependent decision problems; p-value-based rules can silently violate the risk budget (Example 4.10).
- **4.3** Optional continuation: e-values combine multiplicatively across post-hoc-decided follow-up experiments, licensing “living” analyses that grow evidence over time; p-values cannot be safely recombined this way.
- **4.4** Principal-agent testing: incentive compatibility of a statistical contract is exactly captured by requiring scaled license functions to be e-variables – protecting a regulator from agents who already know the truth.
- **Common thread:** in all four settings, the object that remains valid under adversarial, data-dependent manipulation is always the e-value.

- §4.1 combines results explicit or implicit in Wang and Ramdas (2022), Xu et al. (2024), and Koning (2023a), in the context of post-hoc α for testing or FDR/FCR control. The sufficient condition in Remark 4.2 is due to Blanchard and Roquain (2008).
- §4.2 is based on Grünwald (2024), with some presentation differences.
- Grünwald et al. (2024a) coined the term *optional continuation* (§4.3); further explored by Shafer (2023) and Ter Schure and Grünwald (2022) (application to meta-analysis).
- The incentive-compatible principal-agent problem of §4.4 was formulated and studied by Bates et al. (2022).